

DETAILED DESCRIPTION OF THE INVENTION

Commercial Deployment and Licensing Architecture

In various embodiments, the penile anatomical geometry profiling system may be commercially deployed as a standalone device offering geometry profile generation and structural analytics, or as a modular node integrated into a multi-device biometric ecosystem. Deployment may include consumer retail distribution, subscription-based digital services, clinical partnerships, research collaborations, or enterprise licensing arrangements.

In certain implementations, the system may be provided under a modular licensing framework in which geometry acquisition functionality, reconstruction software, encoded profile formats, and comparative modeling engines are licensed individually or in combination. For example, a licensee may obtain rights to manufacture and distribute the hardware device while utilizing licensed reconstruction algorithms or standardized profile schemas for downstream integration. Alternatively, the geometry profiling module may be licensed as a software-defined feature deployed on existing hardware platforms capable of supporting the described sensing architecture.

Commercial deployment may further include sequential integration pathways in which geometry profiling is licensed or distributed as an initial module, followed by integration of additional sensing nodes, calibration engines, or comparative modeling services over time. Such sequential integration may enable staged rollout of functionality while preserving interoperability through standardized geometry profile representations and secure data exchange protocols.

In some embodiments, the system may support enterprise deployment models including white-label hardware distribution, API-based access to encoded geometry profiles, or secure integration with third-party personalization, matching, or device calibration platforms. Access to geometry profile datasets may be governed by licensing terms, authentication controls, and cryptographic protections to preserve confidentiality and controlled use.

The commercial deployment and licensing architectures described herein are illustrative and do not limit the disclosure to any specific business model, distribution channel, contractual structure, or integration pathway. Any commercial implementation that utilizes the disclosed sensing architecture, reconstruction engine, and geometry profile generation methodology falls within the scope of the present disclosure.